Automated Optic Disc and Cup Segmentation Using Mask R-CNN for Glaucoma Assessment

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***Abstract*—Glaucoma, a leading cause of irreversible blindness, requires precise segmentation of the optic disc and optic cup in fundus images for early diagnosis and progression monitoring. This study presents an automated deep-learning-based segmentation approach using Mask R-CNN, trained on the G1020 dataset, which consists of 1,020 fundus images with corresponding labeled masks. The dataset was split into an 80% training set (816 images) and a 20% test set (204 images). The model was fine-tuned with a customized predictor for multi-class segmentation to effectively detect the optic disc and cup regions. Training was conducted using a stochastic gradient descent optimizer with a learning rate scheduler to enhance convergence. The proposed model achieved an average Dice coefficient of 0.9381 for the optic disc and 0.8372 for the optic cup, with corresponding Intersection over Union (IoU) scores of 0.8887 and 0.7319, demonstrating robustness in identifying these critical anatomical structures. The segmentation results show significant potential for clinical applications, aiding ophthalmologists in automated glaucoma assessment. Future work will focus on improving the model's generalizability to diverse datasets and integrating it into real-time clinical decision-making systems.**

***Index Terms*—optic disc segmentation, mask r-cnn, deep Learning, glaucoma detection, G1020 dataset.**

I. Introduction

Glaucoma is a progressive neurodegenerative disease and one of the leading causes of irreversible blindness worldwide, affecting approximately 76 million individuals, with projections estimating an increase to 111.8 million by 2040 [1]. The disease primarily results in optic nerve damage due to elevated intraocular pressure, leading to gradual vision loss if not detected and managed early [2]. Since glaucoma is often asymptomatic in its early stages, early detection and continuous monitoring are essential for preventing vision deterioration.

A key clinical indicator for glaucoma assessment is the optic disc (OD) and optic cup (OC) morphology. The cup-to-disc ratio (CDR), defined as the ratio of the vertical diameter of the optic cup to the vertical diameter of the optic disc, is a crucial biomarker in glaucoma screening [3]. Higher CDR values are strongly associated with glaucomatous progression, necessitating accurate and automated segmentation of OD and OC regions in retinal fundus images.

Traditional OD and OC segmentation methods rely on manual delineation by ophthalmologists, which is time-consuming, prone to inter-observer variability, and infeasible for large-scale screening programs [4]. Automated segmentation methods provide a solution by ensuring consistency, reducing human workload, and enabling early detection of glaucoma in asymptomatic patients.

Deep learning has significantly advanced medical image segmentation, with Convolutional Neural Networks (CNNs) demonstrating strong performance in OD and OC segmentation [5]. Architectures such as U-Net have been widely used due to their ability to capture spatial features efficiently. Sevastopolsky [6] proposed an improved U-Net model for OD and OC segmentation, achieving notable accuracy improvements. However, U-Net-based approaches often struggle with precise boundary delineation, particularly for the optic cup, which has a highly variable shape and low contrast with surrounding tissues [7]. Recent works have explored transformer-based models and generative adversarial networks (GANs) for medical segmentation, but these approaches often require extensive computational resources and large-scale datasets.

Recent advances in deep learning for medical image analysis have introduced transformer-based architectures such as Swin Transformers [8], which leverage self-attention mechanisms to capture long-range dependencies and enhance segmentation performance. Transformer models have demonstrated state-of-the-art accuracy in various medical imaging tasks, including optic disc and cup segmentation [9]. However, despite their advantages in capturing global context, transformers typically require significantly larger training datasets and higher computational resources compared to traditional CNNs. Additionally, their performance on small medical datasets, such as G1020, may not generalize well without extensive pretraining.

Given these considerations, this study proposes a Mask R-CNN-based approach for OD and OC segmentation. Unlike pixel-wise segmentation networks, Mask R-CNN integrates object detection with instance segmentation, enabling precise boundary delineation [10]. This is particularly beneficial for detecting the optic cup, which presents challenges due to its smaller size and indistinct edges. Compared to U-Net, Mask R-CNN allows simultaneous detection and segmentation of multiple objects, providing flexibility in handling varying fundus imaging conditions.

To evaluate our model, we utilize the G1020 dataset, a benchmark dataset for OD and OC segmentation in glaucoma assessment [11]. The dataset consists of 1,020 high-resolution fundus images reflecting real-world variations in optic disc positioning and illumination. The dataset is divided into training and testing sets, following standard practices in medical image segmentation. Future work may explore cross-validation techniques to enhance generalizability.

This study contributes to automated OD and OC segmentation by employing Mask R-CNN with a ResNet-50 backbone and Feature Pyramid Network (FPN). We evaluate our model using Dice coefficient and Intersection over Union (IoU) metrics and discuss its strengths, limitations, and potential applications in clinical settings. Our findings aim to support the development of AI-driven glaucoma screening tools, improving diagnostic accuracy and efficiency in real-world healthcare environments.

II. METHODOLOGY

A. Dataset Description

The G1020 dataset is a publicly available benchmark designed for automated glaucoma detection using retinal fundus images. It comprises 1020 high-resolution fundus images, collected from a private ophthalmology clinic in Kaiserslautern, Germany, over a 12-year period (2005–2017). Unlike many existing datasets that impose strict imaging constraints, G1020 captures natural variations in optic disc positioning, illumination, and image quality, making it a real-world representative dataset for developing robust computer-aided diagnostic (CAD) systems. The images were acquired using a 45-degree field of view following pupil dilation, and each image has been fully anonymized before public release.

This dataset is extensively annotated to support multiple glaucoma-related tasks. Among the 1020 images, 1019 include optic disc (OD) annotations, while 791 provide optic cup (OC) segmentations, reflecting challenges posed by cases where the optic cup is not visible. Additionally, 991 images contain optic disc localization (discLoc), aiding in object detection tasks. The ground truth annotations include optic disc and cup segmentation, vertical cup-to-disc ratio (CDR), and neuroretinal rim thickness measurements in four quadrants (Inferior, Superior, Nasal, and Temporal). These annotations were manually labeled using LabelMe and subsequently verified by an experienced ophthalmologist.

The G1020 dataset is an essential benchmark for developing and evaluating automated methods in glaucoma classification, optic disc and cup segmentation, and large-scale screening applications. Compared to conventional datasets, G1020 offers a more complex and diverse set of fundus images, making it well-suited for assessing model performance in real-world clinical environments. Its extensive annotations and inclusion of images captured under various conditions provide a more comprehensive and realistic evaluation framework.

B. Data Preprocessing

The dataset underwent several preprocessing steps to ensure optimal model performance and training efficiency. First, we implemented a custom PyTorch Dataset class to handle the paired image and annotation data. Each fundus image was loaded and converted to RGB format to ensure consistency across the dataset, as variations in color channels can affect the performance of segmentation algorithms [12]. For the ground truth annotations, we processed the JSON files containing polygon coordinates for both optic disc and cup regions. These annotations were originally created using LabelMe and required conversion into a format suitable for training the Mask R-CNN model. The preprocessing pipeline included several key steps:

1) Image Processing: Each fundus image was loaded using PIL (Python Imaging Library) and converted to RGB format to standardize the input. The images were then transformed into PyTorch tensors with normalized pixel values between 0 and 1 using torchvision's functional transforms.
2) Annotation Processing: The JSON annotations were parsed to extract polygon coordinates for both optic disc and cup regions. These coordinates were converted into two distinct formats required by Mask R-CNN:
   - Bounding Boxes: For each annotated region (disc and cup), we computed the minimum and maximum coordinates along both axes to create bounding boxes, facilitating the localization of these structures during training [13].
   - Segmentation Masks: Segmentation Masks: Binary masks were generated for each region using OpenCV's fillPoly function, enabling pixel-wise classification during model training.
3) Label Encoding: We implemented a multi-class labeling system where background was encoded as class 0, optic disc was encoded as class 1 and optic cup was encoded as class 2.
4) Data Validation: To ensure robust training, we implemented validation checks during preprocessing. Images lacking valid annotations were automatically skipped during training, and a warning system was implemented to flag such cases. This approach helped maintain data quality and prevented training on incomplete or corrupted samples.

For model input, we organized the preprocessed data into a dictionary format containing:

- Normalized image tensors
- Ground truth bounding boxes as floating-point tensors
- Binary segmentation masks as unsigned 8-bit tensors
- Image identifiers for tracking and evaluation purposes

The preprocessing pipeline was designed to be memory-efficient, processing images on-demand during training rather than loading the entire dataset into memory. This approach enabled efficient handling of the large-scale dataset while maintaining training performance.

C. Model Architecture

In this study, we employ Mask R-CNN with ResNet-50 and Feature Pyramid Network (FPN) as the backbone to perform optic disc and optic cup segmentation from fundus images. The Mask R-CNN framework is an extension of Faster R-CNN, designed for instance segmentation by incorporating an additional mask prediction branch, allowing for precise delineation of object boundaries [10]. This approach allows simultaneous object detection and pixel-wise segmentation, which is particularly beneficial in medical imaging applications where precise boundary delineation is crucial.

The backbone of our architecture is ResNet-50, a deep residual network pre-trained on the COCO dataset [14], known for its effectiveness in feature extraction [15]. It is integrated with a Feature Pyramid Network (FPN) to enhance multi-scale feature representation, making it more effective in detecting objects of varying sizes. The Region Proposal Network (RPN) generates candidate bounding boxes, which are then refined through the Region of Interest (RoI) Align module to produce accurate object localization, a key component in improving segmentation accuracy in instance-based tasks [16].

To adapt the model for our task, the classification head of the RoI predictor has been modified to accommodate three classes: background, optic disc, and optic cup. Specifically, the original box predictor in Faster R-CNN is replaced with a Fast R-CNN predictor, where the final classification layer's input features are adjusted to match the new number of classes. Likewise, the mask prediction branch is customized by replacing the mask predictor with a new Mask R-CNN predictor, ensuring that it generates per-class segmentation masks of size (num_classes, 28, 28) for the optic disc and optic cup.

The final model comprises the following key components:

- Backbone: ResNet-50 with FPN for multi-scale feature extraction.
- Region Proposal Network (RPN): Generates object proposals with anchor-based bounding boxes.
- RoI Align: Enhances spatial accuracy by preventing quantization errors from feature pooling.
- Classification Head: Modified Fast R-CNN predictor with three-class output (background, optic disc, optic cup).
- Mask Branch: Customized Mask R-CNN predictor with convolutional layers to output per-class segmentation masks of size (num_classes, 28, 28).

Table I provides a detailed breakdown of the Mask R-CNN architecture, including its key layers and output shapes. The backbone consists of ResNet-50 with an FPN for multi-scale feature extraction, while the region proposal network (RPN) generates object candidates. The RoI Align operation refines object localization before the classification and regression heads make the final predictions. The mask branch contains convolutional layers and upsampling operations that generate per-class segmentation masks, ensuring accurate delineation of the optic disc and optic cup.

TABLE I
MASK R-CNN ARCHITECTURE WITH RESNET-50 FPN

Layer	Type	Output Shape
Backbone: ResNet-50 with Feature Pyramid Network (FPN)		
Conv1	Conv2D	$[C, H, W]$
Pool1	MaxPool2D	$[C, H/2, W/2]$
Res2	Bottleneck Block	$[C, H/4, W/4]$
Res3	Bottleneck Block	$[C, H/8, W/8]$
Res4	Bottleneck Block	$[C, H/16, W/16]$
Res5	Bottleneck Block	$[C, H/32, W/32]$
Region Proposal Network (RPN)		
RPN Conv	Conv2D	$[256, H/32, W/32]$
RPN Classifier	Conv2D	$[A, H/32, W/32]$
RPN Regressor	Conv2D	$[4A, H/32, W/32]$
ROI Align and Box Head		
ROI Align	Pooling	$[1024, 7, 7]$
FC1	Fully Connected	$[1024]$
FC2	Fully Connected	$[1024]$
Box Classifier	Fully Connected	$[N, num\_classes]$
Box Regressor	Fully Connected	$[N, 4]$
Mask Head (Segmentation Branch)		
Mask Conv1	Conv2D	$[256, 14, 14]$
Mask Conv2	Conv2D	$[256, 14, 14]$
Mask Conv3	Conv2D	$[256, 14, 14]$
Mask Conv4	Conv2D	$[256, 14, 14]$
Deconv	Deconv2D	$[256, 28, 28]$
Mask Predictor	Conv2D	$[num\_classes, 28, 28]$

D. Training and Evaluation Metrics

The G1020 dataset was divided into 80% training (816 images) and 20% testing (204 images) to ensure a sufficiently large training set while maintaining a representative test set for evaluation. Since PyTorch's Mask R-CNN implementation does not return validation loss in evaluation mode, a separate validation set was not explicitly included.

The model is trained using a dataset of fundus images, where each image is annotated with both optic disc and optic cup masks. The training process employs stochastic gradient descent (SGD) as the optimizer, with a learning rate of 0.001, a momentum of 0.9, and a weight decay of 0.0005. To prevent overfitting and improve generalization, a step learning rate scheduler is applied, reducing the learning rate by a factor of 0.1 every 3 epochs.

The model is trained for 20 epochs, with each epoch processing mini-batches of 4 images. The total loss for each mini-batch is computed by summing the contributions from multiple loss components, including classification loss, bounding box regression loss, mask loss, objectness loss, and RPN box regression loss. The combined loss is then used to perform backpropagation, where the gradients are computed and used to update the model parameters via the optimizer. Figure 1 illustrates the loss trend over epochs, showing the gradual convergence of the model.

For evaluation, we employ the following metrics to assess the performance of the model: Intersection over Union (IoU) and Dice Coefficient, which are widely used in segmentation tasks to measure overlap accuracy between predicted and ground truth masks [4].

- Intersection over Union (IoU): Measures the overlap between the predicted segmentation mask and the ground-truth mask. It is defined as:

$$\text{IoU} = \frac{|A \cap B|}{|A \cup B|}$$

- Dice Coefficient: Evaluates segmentation accuracy by comparing the predicted masks against the ground-truth annotations. It is given by:

$$\text{Dice} = \frac{2|A \cap B|}{|A| + |B|}$$

where A represents the predicted mask and B represents the ground-truth mask.

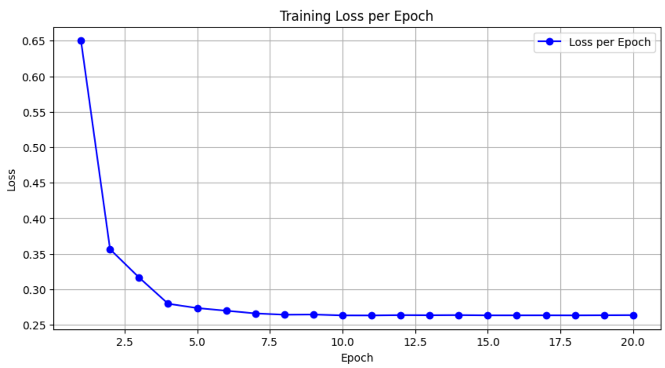

Fig. 1. Training loss curve over 20 epochs, illustrating stable learning behavior with a gradual decrease in loss, indicating effective model convergence.

III. RESULTS

The model was evaluated using a test set consisting of 204 images. Among these, 204 images contained optic disc annotations, while 156 images included optic cup annotations.

A. Overall Evaluation Results

Table II presents the segmentation performance metrics for both the optic disc and optic cup. The model achieved high Dice coefficients of 0.9381 for the optic disc and 0.8372 for the optic cup, with corresponding IoU scores of 0.8887 and 0.7319. These results indicate strong segmentation performance, particularly for the optic disc, demonstrating the model's effectiveness in detecting and delineating key anatomical structures.

TABLE II
SEGMENTATION PERFORMANCE METRICS

Segmentation Task	Average Dice	Average IoU
Optic Disc	0.9381	0.8887
Optic Cup	0.8372	0.7319

Although the model performs well for both segmentation tasks, the relatively lower accuracy for the optic cup can be attributed to its smaller size, lower contrast, and class imbalance in the dataset. Specifically, only 791 out of 1020 images in the G1020 dataset contain optic cup annotations, leading to an uneven distribution of training samples for the two classes.

In clinical applications, segmentation errors in the optic cup can have a direct impact on cup-to-disc ratio (CDR) calculations, which are crucial for glaucoma assessment. Over-estimation or underestimation of the optic cup size can lead to incorrect CDR values, potentially affecting early glaucoma diagnosis. This highlights the importance of accurate segmentation for reliable clinical decision-making.

To address class imbalance, future work could implement weighted loss functions, assigning higher importance to optic cup pixels to improve model optimization. Additionally, data augmentation techniques, such as synthetic sample generation using adversarial learning or oversampling methods, could improve the model's ability to detect optic cups with higher precision.

B. Qualitative Results

To further illustrate the model's segmentation performance, Figure 2. shows a representative example of the ground truth annotation and the corresponding predicted segmentation mask.

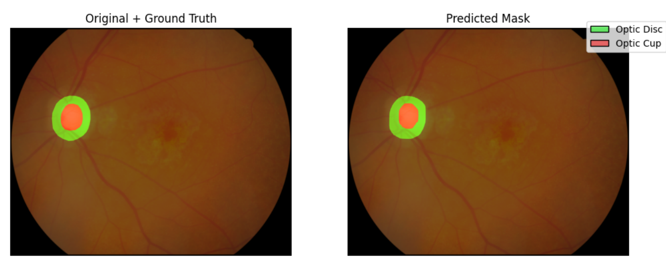

Fig. 2. Qualitative comparison of optic disc and optic cup segmentation results. The left image represents the ground truth annotations, while the right image shows the model's predicted segmentation mask, demonstrating close alignment with expert labels.)

The figure demonstrates that the predicted segmentation masks closely align with the ground truth, confirming the model's effectiveness in delineating the optic disc and optic cup structures.

These quantitative and qualitative evaluations highlight the robustness of the proposed method in segmenting key anatomical structures in fundus images, which is crucial for automated glaucoma assessment.

The results indicate that the proposed Mask R-CNN model performs well in segmenting both the optic disc and optic cup, with higher accuracy for the optic disc. The lower performance for the optic cup can be attributed to the increased variability in cup boundaries and the presence of indistinct edges in some images. The dataset's diverse imaging conditions, including variations in illumination and optic disc positioning, underscore the robustness of our approach.

One limitation observed in our model is the reduced segmentation accuracy for smaller optic cups. This suggests that additional refinement techniques, such as incorporating attention mechanisms or leveraging higher-resolution images, could improve segmentation accuracy. Future work could explore domain adaptation techniques to generalize the model's performance across different fundus imaging datasets.

IV. CONCLUSIONS

In this study, we proposed a Mask R-CNN-based approach for optic disc and optic cup segmentation in fundus images. The model demonstrated high segmentation accuracy, particularly for the optic disc, with an overall Dice score of 0.9381 and IoU of 0.8887. The results show that our method is effective for automated glaucoma assessment, providing reliable segmentation for clinical applications.

Accurate segmentation of the optic disc and cup is a crucial step in the development of fully automated glaucoma detection and classification systems. By ensuring precise boundary delineation, our model lays the foundation for extracting key diagnostic features such as the cup-to-disc ratio (CDR), which is widely used in glaucoma screening. Automated segmentation not only enhances diagnostic consistency but also enables large-scale screening in resource-limited settings where access to ophthalmologists is limited.

Despite its promising performance, there are still areas for improvement. One challenge observed in our study was the relatively lower segmentation accuracy for the optic cup, which may impact the computation of CDR. Future research should focus on enhancing the model's ability to differentiate small and indistinct cup regions, possibly through advanced architectures such as transformer-based models or multi-scale feature aggregation techniques. In conclusion, the proposed approach has demonstrated its potential as a powerful tool for automated optic disc and cup segmentation. With further enhancements in segmentation precision, generalization, and integration into clinical workflows, this model can serve as a foundation for fully automated glaucoma detection and classification systems, ultimately improving early diagnosis and patient outcomes.

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